

Jinpeng Lu

Seeking a Video Generation Research Internship

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M.S. student at USTC, expected Jun. 2028

Available now, 6+ months full-time · Beijing/Shanghai/Hangzhou



Research Profile

Research in **controllable video generation and embodied world models**, focusing on whether video models maintain a consistent world state under changes of viewpoint and shot. Designed and trained camera-trajectory control and multi-shot video generation models on Wan2.2, with hands-on experience in DiTs, flow matching, and distributed training; built a multi-source camera-trajectory video data pipeline; proposed **WRBench** as first author, a benchmark for persistent event state (23 models, 9,600 videos), with a VLM-based automatic scorer and human evaluation; first-author papers at **ICLR 2026 and MICCAI 2024**.

Education

University of Science and Technology of China

M.S. in Information and Communication Engineering; Advisor: Prof. Zhiwei Xiong

Hefei

Sep. 2025 – Jun. 2028 (expected)

Harbin Institute of Technology, Shenzhen

B.S. in Data Science and Big Data Technology; GPA 3.728/4, rank 17/172 (top 10%)

Shenzhen

Sep. 2021 – Jun. 2025

Internship Experience

Beijing Innovation Center of Humanoid Robotics (X-Humanoid, UBTECH)

Embodied World Model Team, Algorithm Intern (Large Models)

Beijing

Nov. 2025 – Sep. 2026

- **Independently owned two research lines**, camera-controllable video generation and few-step sampling for action policies, covering method design, model training, and evaluation; for a Cosmos3-Edge action policy, one-step sampling was non-inferior to 30 steps on LIBERO-10 (three runs of 500 episodes, 95–96% vs. about 93%) while median server-side latency fell from 697 ms to 141 ms.
- **Proposed WRBench as first author** to test whether camera motion alters generated events; built the benchmark design, evaluation system, and paper.
- Located a silent defect in open-source Cosmos-Predict2.5: the action encoder was built after weight initialization and stayed all-zero, so four different action inputs produced bit-identical outputs; the fix restored action responses. Built a controllability evaluation for action-conditioned video models with zero/reversed/shuffled-action controls, bootstrap confidence intervals, and metric-floor audits, and evaluated and selected video foundation models.
- **Collaborative embodied research**: contributed to the core design and refinement of **VLA-Trace** (EMNLP 2026); mainly handled data processing for **Pelican-Unify 1.0**, supporting joint generation of future video and actions.

Research Experience

Camera-Controllable Video Generation for Embodied World Models

Dec. 2025 – Mar. 2026

- **Problem and motivation**: Embodied world models must synthesize new views as a robot turns or moves and keep scenes and events consistent when the view leaves and returns. Embodied data is almost entirely fixed-camera and unposed, so transferred camera-control methods tend to follow text or memorized samples rather than the camera, and camera motion corrupts arm dynamics.
- **Method**: On Wan2.2-TI2V-5B, systematically compared two camera conditions, input-level Plücker rays through a convolutional adapter (CameraCtrl-style) and layer-wise relative-pose injection (ReCamMaster-style), together with staged training and frozen-backbone vs. full fine-tuning; to remove text shortcuts, cleaned camera-motion phrases from about 2.7M captions with an LLM.
- **Data**: **Built a multi-source camera-trajectory data pipeline** (MultiCamVideo, SpatialVID, DynPose, RealEstate10K, etc.), unifying coordinates, intrinsics, and frame rates, fixing sparse-pose interpolation with SLERP, completing unposed videos with ViPE, and balancing sampling by motion; diagnosed biases such as about 99% forward motion in SpatialVID, and wrote a phone-LiDAR capture protocol.
- **Findings**: (1) Camera-motion words in captions and pose leakage create text shortcuts that disable the camera condition; (2) controllability trades off against embodied dynamics: synthetic multi-view data alone gives strong control but loses arm dynamics, while mixing in embodied data keeps dynamics but clearly weakens control; (3) mixing sources is not monotonically helpful, with different datasets causing jitter, text-following, and capture-trajectory-following failures; (4) a moving view does not show that events were preserved, which led to WRBench.

WRBench: Evaluating Persistent Event State in Video World Models

Feb. – Jul. 2026

- **Problem and motivation**: Video generators are increasingly called world models, yet plausible frames do not mean a world that keeps evolving: camera motion should change only the view, not the event. Existing benchmarks cover visual quality, physical plausibility,

and camera control, but not whether a model keeps the same evolving world when observation changes.

- **Method:** Modeled camera motion as a **camera intervention** that changes only observation, so the target is visible, leaves view, and returns; used a 2×2 **event design crossing spatial relocation and state change** over 25 scene families, and staged the evaluation into camera execution, visible consistency, re-observation support, and reobserved endpoint correctness, separating “not seen again” from “seen but wrong”; mapped one intervention onto four native control interfaces.
- **Evaluation system:** Built an evaluation-data pipeline (LLM scene taxonomy, text-to-image first frames, image-to-video prompts) with leakage checks; designed a VLM scorer (yes/no token probabilities, paired positive and negative questions for debiasing) alongside VGGT- Ω camera execution and DINOv2 visual integrity; wrote single-video rating and pairwise blind-comparison protocols, developed a web annotation tool, and checked the evaluators with over 4,100 human judgments (Spearman ρ 0.54–0.71).
- **Findings:** Across **23 models and 9,600 videos**: (1) camera changes leak into event evolution, as merely reversing the turn direction shifts re-observation support by 17.8–51.3 points; (2) state change lowers visible spatial consistency by 13.0 points in every configuration; (3) models most often re-observed do not lead on endpoint correctness; (4) scale and version upgrades mainly improve what is seen rather than what persists. Persistent event state should therefore be a separate evaluation and training target.

Multi-Shot Video Generation: Entity Persistence and State Updates across Shots

May – Sep. 2026

- **Problem and motivation:** Multi-shot storytelling must both persist and update: characters and scenes must persist across shots, while camera, action, and event state change as scripted (a hat taken off should not reappear). Joint generation is window-bounded, and shot-by-shot continuation loses identity and reverts state after hard cuts.
- **Method:** On Wan2.2-TI2V-5B, explored two routes: story and entity semantic tokens denoised jointly with video latents (**joint semantic–video flow matching**), so persistence and updates share one objective; and reference-frame memory that feeds previous-shot keyframes as clean latent prefixes through the native first-frame channel (adopting StoryMem’s negative RoPE shift), conditioned on post-event Keep/Change state captions.
- **Evaluation and data:** Built 184K-story multi-shot training data with quality scores and shot-boundary labels for full-parameter training; designed a script-compiled state benchmark (100 hard-cut scripts, 2,000+ assertions), validated the judge on synthetic failures, and judged outcomes with story-clustered bootstrap CIs and preregistered gates.
- **Diagnostic findings:** Controlled probes show that reference-frame memory triggers carry-over of previous-shot content when the memory is self-generated and the text matches (+29.5 points over a no-memory control, CI excluding 0); common cross-shot consistency metrics reward copying and missing cuts (real videos score below most generated ones), so evaluation moved to script assertions. Research ongoing.

Publications

ICLR 2026 CCF-A	Johnson-Lindenstrauss Lemma Guided Network for Efficient 3D Medical Segmentation Jinpeng Lu , Linghan Cai, Yinda Chen, Guo Tang, Songhan Jiang, Haoyuan Shi, Zhiwei Xiong Proposed JL lemma-guided grouped convolution and Paired Window Attention, and transferred texture knowledge from a self-supervised teacher through channel Gram matrices, yielding the 1.66M-parameter VeloxSeg.
arXiv 2026 WRBench	Current World Models Lack a Persistent State Core Jinpeng Lu , Dexu Zhu, Haoyuan Shi, Linghan Cai, Guo Tang, Yinda Chen, Jie Cao, Duyu Tang, Yi Zhang, Yong Dai, Xiaozhu Ju Introduced WRBench for persistent event state, finding that current video models improve how a world is rendered and revisited faster than how its evolving state persists.
Tech Report 2025 Co-first author	Does DINOv3 Set a New Medical Vision Standard? Che Liu, Yinda Chen, Haoyuan Shi, Jinpeng Lu , et al. Benchmarked how natural-image pretraining transfers to medical 2D/3D tasks.
MICCAI 2024 CCF-B Early Accept	H2ASeg: Hierarchical Adaptive Interaction and Weighting Network for Tumor Segmentation in PET/CT Images Jinpeng Lu , Jingyun Chen, Linghan Cai, Songhan Jiang, Yongbing Zhang Designed modality-cooperative attention and target-aware weighting to fuse PET and CT.

Service and Skills

Academic Service	Reviewer for ICLR 2027, NeurIPS 2026, and IEEE TNNLS.
Generation and Control	DiT, flow matching, video diffusion fine-tuning (Wan2.2), camera conditioning (Plücker rays, relative poses), reference-frame conditioning, joint video–semantic modeling, knowledge distillation.
Data and Evaluation	Multi-source video data construction, camera pose estimation (ViPE, VGGT), coordinate and trajectory normalization, LLM-based caption cleaning; VLM-based automatic scoring, human evaluation protocols and annotation tools, confidence intervals and paired statistical tests.
Training and Engineering	Python, PyTorch, DeepSpeed ZeRO, multi-node distributed training and full-parameter fine-tuning, checkpoint resumption, inference profiling.